

Unified Enterprise AI Agents Platform

Architecture Documentation

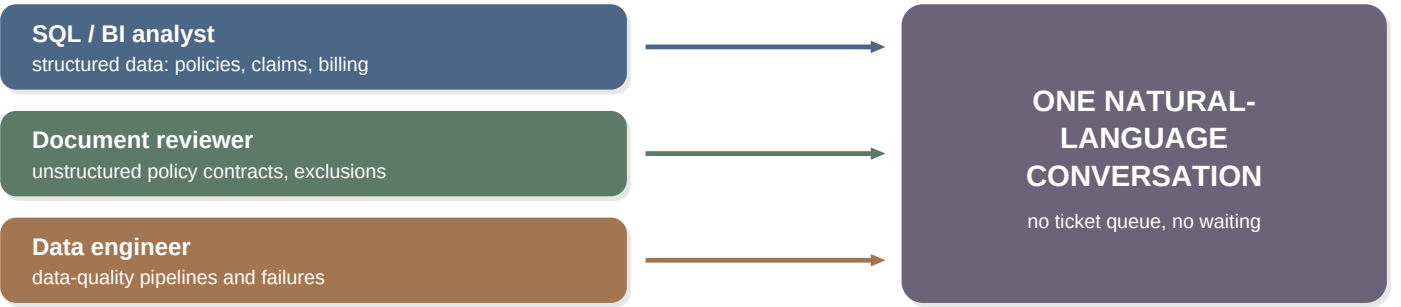
Built natively on Snowflake Cortex | Insurance self-service analytics, documents and data quality

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Problem

Problem statement: Insurance business users need self-service answers from three disconnected sources: structured operational data, unstructured policy documents, and data-quality pipelines, without waiting on a SQL analyst, document reviewer, or data engineer each time.

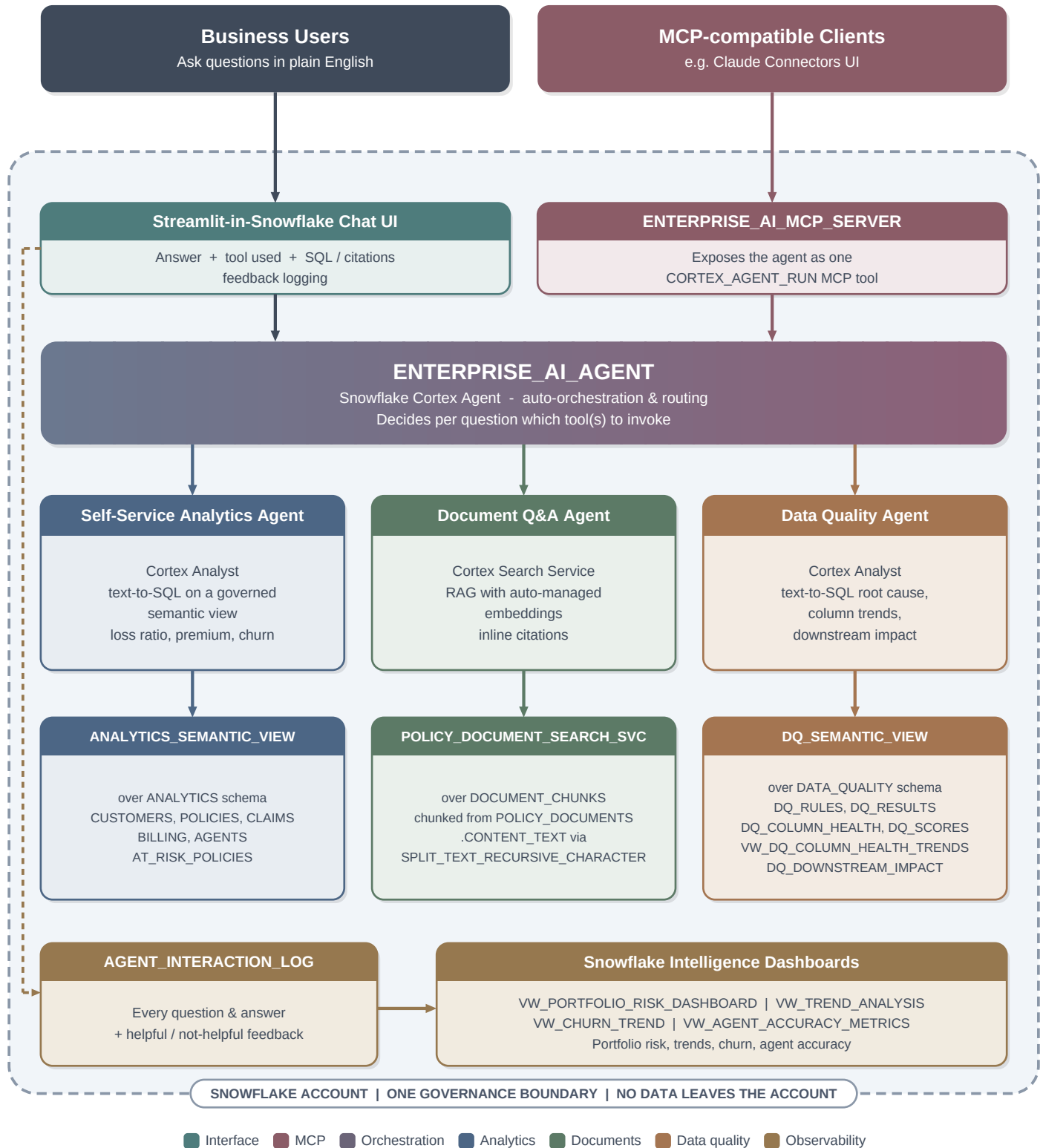
Business users at insurers need answers from structured operational data (policies, claims, billing), unstructured documents (policy contracts, exclusion clauses), and data-quality pipelines, but today each requires a different specialist. This platform collapses all three into one natural-language conversation.



Three capabilities at a glance

Capability	Engine	Grounded in
Self-Service Analytics	Cortex Analyst (text-to-SQL)	ANALYTICS_SEMANTIC_VIEW policies, claims, billing, agents
Document Q&A	Cortex Search Service (RAG)	POLICY_DOCUMENT_SEARCH_SVC policy contracts, exclusion clauses
Data Quality Root Cause	Cortex Analyst (text-to-SQL)	DQ_SEMANTIC_VIEW rules, results, scores, lineage

2 High-level architecture



Everything runs natively inside Snowflake: one platform, one governance boundary, no data leaves the account, and no external vector database or orchestration service is required. The agent runtime sanitizes spaces and ampersands in tool names it returns (logs, traces): Self-Service_Analytics_Agent, Document_Q_A_Agent, Data_Quality_Agent.

3 Component design

AGENT 1	Self-Service Analytics A native SEMANTIC VIEW (ANALYTICS_SEMANTIC_VIEW) declares business-friendly dimensions, facts, and metrics (loss ratio, total premium, fraud claim count, revenue at risk, churn probability) over six operational tables, with relationships declared explicitly so Cortex Analyst always joins correctly. Cortex Analyst translates a natural-language question into governed SQL against this view.
AGENT 2	Document Q&A (RAG) POLICY_DOCUMENTS.CONTENT_TEXT is chunked with SPLIT_TEXT_RECURSIVE_CHARACTER into DOCUMENT_CHUNKS. A CORTEX SEARCH SERVICE indexes CHUNK_TEXT, with POLICY_ID, DOCUMENT_TYPE, and DOCUMENT_TITLE as filterable attributes. Cortex Search generates and manages embeddings internally, so there is no separate embedding pipeline to maintain.
AGENT 3	Data Quality Root Cause A second semantic view (DQ_SEMANTIC_VIEW) sits over DQ_RULES, DQ_RESULTS, DQ_COLUMN_HEALTH, DQ_SCORES, VW_DQ_COLUMN_HEALTH_TRENDS (per-column trend/delta), and DQ_DOWNSTREAM_IMPACT (lineage to affected dashboards and reports). Users ask "why did X fail", "which column dropped the most", or "is any downstream reporting impacted" and get an evidence-backed answer.
ORCHESTRATION	One agent, three tools All three are registered as tool_resources on one AGENT object (ENTERPRISE_AI_AGENT). The Cortex Agents orchestration model decides per question which tool(s) to invoke. The "unified" requirement is met at the platform level, not by the user having to pick an agent.
FRONT END	Streamlit-in-Snowflake chat Calls agents/ENTERPRISE_AI_AGENT:run via _snowflake.send_snow_api_request, renders the answer, shows which tool answered, and expands the generated SQL or document citations inline for transparency. Every question and answer is logged to AGENT_INTERACTION_LOG, including helpful / not-helpful feedback.
DASHBOARDS	Snowflake Intelligence dashboards A second Streamlit page reads three view groups: VW_PORTFOLIO_RISK_DASHBOARD (premium, loss ratio, revenue at risk by policy type and region), VW_TREND_ANALYSIS / VW_CHURN_TREND (month-over-month claims, fraud, churn), and VW_AGENT_ACCURACY_METRICS (query volume and helpful-rate per tool).
MCP	MCP integration ENTERPRISE_AI_MCP_SERVER (CREATE MCP SERVER) exposes the agent itself as a single CORTEX_AGENT_RUN MCP tool, so any MCP-compatible client (tested live with Claude's Connectors UI) gets the same fully executed, orchestrated answers as the Streamlit app, not just generated SQL.

4 Benefits

Faster answers Questions that took hours (waiting on an analyst, a document review, or a DQ ticket) now resolve in seconds, in one conversation.	No new tools Business users ask in plain English: no SQL, no BI dashboard navigation, no ticket queue.	Trustworthy by design Every answer shows its generated SQL or source document citation, so it can be verified, not just trusted.	Many channels A single Cortex Agent and its MCP server serve Streamlit and any MCP-compatible client, without separate integrations per capability.
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